

Ossp skill(week -2)

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1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    while (1)
    {
        // Display shell prompt
        printf("myshell$ ");
        fflush(stdout);
```

```

// Read user input
if (fgets(input, sizeof(input), stdin) == NULL)
{
    printf("\n");
    break;
}

// Remove newline
input[strcspn(input, "\n")] = '\0';

// Ignore empty input
if (strlen(input) == 0)
    continue;

// Handle exit command
if (strcmp(input, "exit") == 0)
{
    printf("Exiting shell...\n");
    break;
}

// Split input into arguments
int argc = 0;
char *token = strtok(input, " ");

while (token != NULL && argc < MAX_ARGS - 1)
{
    args[argc++] = token;
    token = strtok(NULL, " ");
}

args[argc] = NULL;

```

```
// Built-in cd command
if (strcmp(args[0], "cd") == 0)
{
    if (args[1] == NULL)
    {
        fprintf(stderr, "cd: missing directory\n");
    }
    else
    {
        if (chdir(args[1]) != 0)
            perror("cd");
    }

    continue;
}
```

```
// Create child process
pid_t pid = fork();
```

```
if (pid < 0)
{
    perror("fork");
    continue;
}
```

```
// Child process
if (pid == 0)
{
    execvp(args[0], args);
```

```
// If execvp fails
```

```

        perror("Command failed");
        exit(EXIT_FAILURE);
    }

    // Parent process
    else
    {
        waitpid(pid, NULL, 0);
    }
}

return 0;
}

```

Output:

```

aruntej@LAPTOP-DGRT8C5G:~$ nano skill-2.c
aruntej@LAPTOP-DGRT8C5G:~$ gcc skill-2.c -o skill-2
aruntej@LAPTOP-DGRT8C5G:~$ ./skill-2
myshell$ pwd
/home/aruntej
myshell$ ls
Hello.c          exp3             experiment-2     experiment-4.c  notes.txt       program1
destination.txt  exp3.c          experiment-2.c  experiment.c    ossp            program1.c
exp1.c           experiment       experiment-3     fork            output.txt      program2
exp2             experiment-1     experiment-3.c  fork.c          producerconsumer program2.c
exp2.c           experiment-1.c  experiment-4    input.txt       producerconsumer.c program3
myshell$ date
Tue Aug 25 02:54:26 UTC 2026
myshell$ whoami
aruntej
myshell$ cd
cd: missing directory
myshell$ |

```